

EDUCATION

- **University of Alberta** Edmonton, AB
PhD in Computing Science; GPA: 4.0/4.0 2024 – Present
 - **Supervisor:** Dr. Levi Lelis
 - **Research Interests:** LLM World Modeling, Reinforcement Learning, Programmatic Policy Synthesis, Option Discovery and Hierarchical RL
- **University of Alberta** Edmonton, AB
MSc in Computing Science; GPA: 4.0/4.0 2019 – 2022
 - **Supervisor:** Dr. Martin Müller
 - **Thesis:** Monte Carlo Tree Search in the Presence of Model Uncertainty, which led to publication [C3].
- **Iran University of Science and Technology** Tehran, Iran
BSc in Computer Engineering; GPA: 3.87/4.0 2015 – 2019
 - **Supervisors:** Dr. Nasser Mozayani and Dr. Sauleh Eetemadi
 - **Thesis:** Image Captioning using Attention Mechanisms for the Farsi Language.

PUBLICATIONS

Conference Papers

- [C6] NEURIPS, 2026: Parnian Behdin, Reza Abdollahzadeh, **Kiarash Aghakasiri**, Levi Lelis. *Learning Reusable Options by Decomposing Neural Policies*
- [C5] ICML, 2026: Amirhossein Rajabpour, **Kiarash Aghakasiri**, Sandra Zilles, Levi Lelis. *Revisiting OOD Generalization in Programmatic RL*
- [C4] WACV, 2026: Bahador Rashidi, **Kiarash Aghakasiri**, Shupeiz Zhang, Amirmohsen Sattarifard, Yue Zhang, Chao Gao. *ScoreNet: Netting Lightweight Quality Scores for Better Visual Assessment with Large Multi-Modality Models*
- [C3] AAAI, 2024: Farnaz Kohankhaki*, **Kiarash Aghakasiri***, Hongming Zhang, Ting-Han Wei, Chao Gao, Martin Müller. *Monte Carlo Tree Search in the Presence of Transition Uncertainty*
- [C2] ICASSP, 2024: Bahador Rashidi, **Kiarash Aghakasiri**, Chao Gao, Shuting Zhang, Yue Zhang, Ying Liu, Fengyu Sun. *A Multi-Scale Objective Function for Camera Color Correction*
- [C1] DATE, 2024: Bahador Rashidi, Shan Lu, **Kiarash Aghakasiri**, Chao Gao, Fred Xuefei Han, Zhisheng Wang, Laiyuan Gong, Fengyu Sun. *CASCO: Cascaded Co-Optimization for Neural Network Acceleration*

Journal Papers

- [J1] JOURNAL OF PSYCHOLINGUISTIC RESEARCH, 2021: Fatemeh Karimkhani, Hossein Rahmani, Arezoo Zare, **Kiarash Aghakasiri**, Raana Saheb Nassagh. *Tarvajeh: Word Association Norms for Persian Words*

RESEARCH EXPERIENCE

- **University of Alberta** Edmonton, AB, Canada
PhD Research Assistant 2024 – Present
 - **Didec:** Contributed to the implementation of DIDEc, a neural option-discovery method based on policy masking, which led to publication [C6].
 - **Programmatic vs. Neural Policies:** Contributed to a study re-evaluating prior claims that programmatic RL policies generalize better than neural policies on out-of-distribution tasks; results published as [C5].
- **Huawei** Edmonton, AB, Canada
Support Researcher 2022 – 2024

- **Multi-Modal Image Quality Assessment:** Designed and implemented a new image quality assessment metric leveraging multi-modal foundation models (CLIP-IQA, Q-ALIGN), achieving performance surpassing state-of-the-art baselines. This work led to publication [C4].
- **ISP Pipeline Optimization:** Applied genetic algorithms, sample-efficient Bayesian optimization, and reinforcement learning methods to optimize the image signal processing (ISP) pipeline for edge devices.
- **Color Metric for Human Perception:** Developed a human-aligned color correction metric addressing inconsistencies in ΔE , resulting in publication [C2]. Applied this metric to the edge-device color correction pipeline and improved performance by over 20%.
- **Tensor Optimization:** Implemented a fusion-aware component within the tensor optimization pipeline, improving the previous pipeline by at least 10% and contributing directly to publication [C1].
- **Compiler Optimization:** Applied the novel GMCTS algorithm to compiler optimization tasks from COMPILERGYM.

• Huawei

Edmonton, AB, Canada

Research Intern

Summer 2021

- **Heteroscedastic Regression:** Developed a heteroscedastic regression model for Halide tensor optimization, enabling improved handling of model uncertainty.

• University of Alberta

Edmonton, AB

MSc Research Assistant

2019 – 2022

- **UA-MCTS:** Developed a novel method (UA-MCTS) to adapt vanilla MCTS to transition model inaccuracies, which led to publication [C3].
- **Additional Projects:** Contributed to several exploratory ML/RL studies, including modeling uncertainty via heteroscedastic regression, analyzing TD(λ) stability, improving variance reduction in off-policy evaluation, and investigating out-of-vocabulary handling in sequence tagging models.

TEACHING EXPERIENCE

- **University of Alberta:** TA for Foundations of Computation I (2019, 2020, 2024); Basics of Machine Learning (2020, 2021, 2025).
- **Iran University of Science and Technology:** TA for AI (2018), NLP (2019), Computational Intelligence (2019), Theory of Automata (2017), Foundations of Programming (2016).

TECHNICAL SKILLS

Research Areas: Reinforcement Learning, Hierarchical Reinforcement Learning, Monte Carlo Tree Search, Option Discovery, Program Synthesis, Foundation Models, World Modeling, Bayesian Optimization, Computer Vision, Image Quality Assessment.

Machine Learning: PyTorch, TensorFlow, Ray RLlib, Optuna, Weights & Biases, Scikit-learn.

Programming Languages: Python, C++, Bash.

Systems & Tools: Linux, Docker, Git, Slurm, Jupyter.

HONORS & AWARDS

- **Recruitment Award:** Granted the Recruitment Award valued at \$5,000 CAD by the Department of Computing Science, University of Alberta (2024).
- **2nd-ranked BSc Graduate:** Graduated as the 2nd top student (BSc) from the Department of Computer Engineering at Iran University of Science and Technology (IUST) (2019).
- **National Grad School Examination:** Ranked 1,170 among 181,000 contestants in the Iranian National Graduate School Entrance Examination (2015).
- **NODET Member:** Member of the National Organization for Development of Exceptional Talent (NODET) (2010–present).